MARSHALL RASENDRIA MAHENDRA

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PROFILE

AI/ML Engineer, currently an Information Systems undergraduate at Telkom University. Passionate about building intelligent, real-world applications, from training and deploying machine learning models to developing the full-stack web platforms that bring them to life. Holds 5 professional certifications from Google, IBM, Microsoft, and HackerRank, with a proven track record of shipping 6+ end-to-end products and reaching national finalist stages in 3 hackathons within the past 6 months.

EDUCATION

Telkom University | Bachelor of Information Systems — August 2023 – Present

- Current GPA: 3.69/4.00
- Relevant Coursework: Web Application Development, Object-Oriented Programming, Algorithm and Programming, Data Mining, Cloud Computing, Cyber Security, and Enterprise Resource Planning.
- Involvements: Active Member of GDGoC Machine Learning Study Group across 12+ weekly sessions (exploring Python, TensorFlow, and computer vision implementation).

TECHNICAL SKILLS & COMPETENCIES

- Microsoft Power Platform: Power Apps, Power Automate, AI Builder, TypeScript Office Scripts, approval workflows, and Microsoft 365 integration.
- Artificial Intelligence & Machine Learning: Python, PyTorch, TensorFlow, Scikit-Learn, OpenCV, transfer learning, model evaluation, and explainability.
- LLM & Document Intelligence: OpenAI API, LangChain, RAG, prompt engineering, OCR, invoice data extraction, and human-in-the-loop validation.
- Enterprise Process Automation: SAP S/4HANA invoice data mapping, posting-template alignment, Accounts Payable workflows, duplicate detection, and audit logging.
- Data & Backend Engineering: SQL, Pandas, NumPy, FastAPI, RESTful APIs, PostgreSQL, Docker, and Git/GitHub.
- AWS & Generative AI Fundamentals: Amazon Bedrock, foundation models, and agentic AI concepts.

WORK EXPERIENCES

Artificial Intelligence Engineer | PT Bumi Resources Tbk (Internship) — August 2026 – Present

- Developing AI-powered Accounts Payable automation using Microsoft Power Automate, AI Builder, and TypeScript Office Scripts, transforming invoice documents into structured financial data aligned with SAP S/4HANA posting requirements.
- Translating finance and SAP requirements into automated data mapping across a 43-column posting template, including document headers, invoice line items, currencies, and debit-credit validation.
- Building human review and approval workflows with duplicate detection, automated notifications, and audit logging, while refining AI extraction prompts and validating outputs against sample invoices.

Artificial Intelligence Engineer | PT Data Labs Analytics (Internship) — June 2026 – August 2026

- Delivering AI/ML solutions for an enterprise client as part of the company's outsourced engineering team, contributing across the full model development lifecycle.
- Operating within a client-facing engagement under confidentiality constraints, translating business requirements into technical solutions while following the team's production standards for reproducibility, code quality, and reliable delivery.
- Gained hands-on experience working within the Google Cloud Platform environment, using BigQuery for large-scale data querying and processing as part of the client's cloud-based data workflow.

Artificial Intelligence Engineer | HUMIC Engineering Research Center (Internship) — April 2026 – June 2026

- Designed and implemented reproducible mammogram classification pipelines, covering dataset curation, image preprocessing, patient-level splitting, and clinical metadata encoding.
- Developed and trained six deep learning architectures across whole-image and lesion-region inputs, including single-backbone, hybrid attention, multimodal, and mammography foundation models.
- Configured controlled training and evaluation workflows using staged fine-tuning, mixed precision, gradient accumulation, and validation-based checkpoint and threshold selection.

ORGANIZATIONAL EXPERIENCES

Full-Stack Developer | Central Computer Improvement (CCI) Telkom University — February 2026 – Present

- Served as backend developer on a 6-person engineering team building a vendor marketplace platform in partnership with UKM CSC, developing 90+ RESTful API endpoints using NestJS, PostgreSQL, and Prisma ORM. Authored full API documentation, tested endpoints with Postman, and configured Supabase, Midtrans payment gateway, SMTP, and Google OAuth integrations.

- Independently built a full stack document automation platform for Telkom University's student directorate, enabling 50+ student organizations to auto generate Proposal Kegiatan and LPJ documents, cutting processing time from 2+ hours to minutes. Designed the system from scratch with no predefined spec, replicated official templates via dompdf, and passed a penetration test from Telkom's PuTI Directorate before deployment.

Laboratory Assistant | ERP Laboratory Telkom University March 2026 – Present

- Lead the Champions Teams work program, an initiative aimed at increasing competitive participation among 200+ Information Systems students through curated competition opportunities.
- Research, curate, and disseminate 10+ national and international competitions every month across Essay, Business Model Canvas, Design, IT, and Business Plan categories.

PROJECTS

KawanTuli | Capstone Project July 2026 - Present

- Developing KawanTuli, a BISINDO (Indonesian Sign Language) accessibility platform combining a two-way translator with an avatar-based learning system, built with Next.js, TypeScript, and TensorFlow.js, with all camera inference running on-device so no video ever leaves the browser.
- Trained and deployed two recognition models: a 26-class alphabet classifier at 99.31% validation accuracy (~9,700 images), and a 32-class word-level 1D CNN at 83.13% signer-independent accuracy (1,600 videos), improved from 75.31% by fusing MediaPipe hand landmarks with arm pose features, running in real time at 15 fps with 6 ms median latency.
- Built a rigged 3D avatar (react-three-fiber) that performs sign poses derived from dataset landmarks via analytic IK, plus two internal authoring tools enabling non-engineers to correct poses and re-record gestures via webcam.

AMANIN | Social Project April 2026 – May 2026

- Developed AMANIN, an AI-powered image and video detection app designed to prevent media-based fraud by leveraging self-built deepfake detection models and digital forensic techniques, building the entire web application independently from end to end.
- Trained a custom deepfake detection model using PyTorch and OpenCV on a dataset of 10,000+ images and videos, achieving 90%+ detection accuracy in classifying AI-generated and manipulated media.
- Delivered as a social impact initiative educating 40+ residents across 1 rural village, raising community awareness of the dangers of fraud using AI-generated images and videos and equipping them to recognize manipulated media.

MLCRAFT | Open-Source AI/ML Engineering Plugin July 2026

- Built mlcraft, an open-source plugin that turns AI coding assistants (Claude Code, Antigravity) into a senior AI/ML research engineer, packaging 23 skills that auto-route a plain-language task to the best method across 14 domains including computer vision, medical imaging, NLP and LLM, tabular, time-series, generative, reinforcement learning, graph, multimodal, video, and 3D.
- Encoded a research-first, leakage-aware workflow grounded in 2021 to 2026 literature: paper review with realistic-ceiling estimation, group, patient, and temporal split audits, per-problem metric and explainability selection (ROC-AUC vs mAP vs RMSE, Grad-CAM vs bounding boxes vs SHAP), and an iterative accuracy-improvement loop.

KKA-DRAFT | Telkom University Student Directorate April 2026 – June 2026

- Internal tool for Telkom University's student directorate enabling UKMs and student organizations to generate official Proposal Kegiatan and LPJ documents without the chaos of manual formatting.
- Delivered under a 3 month self-imposed sprint with full creative freedom and no predefined specification, with every design decision, feature, and UI flow owned end-to-end from scratch.
- Solved complex PDF rendering with dompdf including multi-page layouts, embedded tables, signatures, logos, and pixel-accurate formatting, replacing a previously manual process used across 50+ campus organizations.

PUBLICATION

Controlled Comparison of Multimodal Hybrid Deep Learning for Benign-Malignant Mammogram Classification on CBIS-DDSM

First Author | JADIMI: Journal of AI-Driven Informatics and Management Information August 2026

- Identified clinical metadata fusion as the largest source of performance improvement, increasing hybrid-model test AUC from 0.797 to 0.856 on lesion regions and 0.785 to 0.845 on whole mammograms.
- Developed the study's highest-AUC ROI configuration by combining EfficientNet-B3, ConvNeXt-Tiny, gated fusion, CBAM attention, and seven clinical descriptors under a controlled evaluation protocol.

HONORS & AWARDS

Finalist July 2026

OpenAI Build Week Hackathon 2026 | OpenAI

Finalist May 2026

Refactory Hackathon 2026 | Refactory

Finalist April 2026
Hackfest 2026 | Universitas Ciputra

CERTIFICATIONS

- AWS Certified AI Practitioner
  Issued: October 2026 | Credential: Credly
- Google Advanced Data Analytics Professional | Google
  Issued: July 2026 | Credential ID: NZH3N1FVRQVS
- SQL (Advanced) | HackerRank
  Issued: July 2026 | Credential ID: 4fb5ddcd42d8
- Google AI Professional Certificate | Google
  Issued: March 2026 | Credential ID: WZVZJWMHBSNY
- IBM AI Developer Professional Certificate | IBM
  Issued: March 2026 | Credential ID: QT3HO3K513XT
- Microsoft Certified: Azure AI Fundamentals | Microsoft
  Issued: September 2025 | Credential: Credly

LANGUAGES

- Indonesian: Native / Bilingual Proficiency.
- English: Professional Working Proficiency.